

Naive Bayes's Algorithm (Classification)

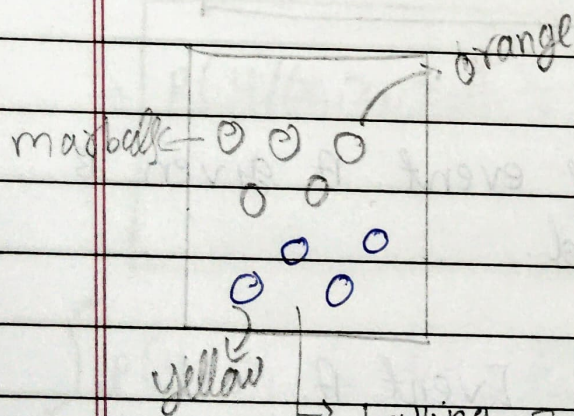
- ① Probability
- ② Bayes's theorem

⇒ Independent Event

Rolling a Dice $\Rightarrow \{1, 2, 3, 4, 5, 6\} \leftarrow$ outcomes.

$$Pr(1) = \frac{1}{6} = Pr(2) = Pr(3) = \dots = Pr(6)$$

⇒ Dependent Event



① what is the probability of removing a orange marble and then a yellow.

$$\left. \begin{array}{l} \text{taking} \\ \text{orange} \\ \text{marble} \end{array} \right\} \Rightarrow \frac{\text{Possible}}{\text{total}} = \frac{1}{9} \rightarrow \text{first event}$$

after taking out the orange marble the total marble is 8

$$\text{So taking yellow} = \frac{\text{Possible}}{\text{total}} = \frac{1}{8} \rightarrow \text{2nd event}$$

conditional probability

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Baye's theorem.

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) * P(B/A) = P(B) * P(A/B)$$

Baye's theorem $\Rightarrow$

$$P(A/B) = \frac{P(A) * P(B/A)}{P(B)}$$



(i) $P(A/B) \Rightarrow$ Probability of event A given B has occurred.

(ii) $P(A) \Rightarrow$ Probability of Event A.

(iii) $P(B) \Rightarrow$ Probability of Event B.

(iv) $P(B/A) \Rightarrow$ Probability of Event B given A has occurred.

Input / Independent

Data

x_1	x_2	x_3	y	Output/ Dependent
—	—	—	yes	
—	—	—	no	
—	—	—	yes	
—	—	—	no	

So if we want to calculate or finding the value of output then we have write it like this: $\downarrow$

$$P(y / (x_1, x_2, x_3))$$

$$\Rightarrow P(y / (x_1, x_2, x_3)) = \frac{P(y) * P(x_1, x_2, x_3 / y)}{P(x_1, x_2, x_3)}$$

 $\Downarrow$

$$P(y / (x_1, x_2, x_3)) = \frac{P(y) * P(x_1 / y) * P(x_2 / y) * P(x_3 / y)}{P(x_1) * P(x_2) * P(x_3)}$$

$$\Rightarrow P(\text{yes} / (x_1, x_2, x_3)) = \frac{P(\text{yes}) * P(x_1 / \text{yes}) * P(x_2 / \text{yes}) * P(x_3 / \text{yes})}{P(x_1) * P(x_2) * P(x_3)}$$

$$P(N_0/x_1, x_2, x_3) = \frac{P(N_0) \cdot P(x_1/N_0) \cdot P(x_2/N_0) \cdot P(x_3/N_0)}{P(x_1) \cdot P(x_2) \cdot P(x_3)}$$

Let's take an example e-① Outlook

weather	yes	no	$P(E/\text{yes})$	$P(E/\text{no})$
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5

① Temperature

Temp	yes	no	$P(E/\text{yes})$	$P(E/\text{no})$
Hot	2	2	2/9	2/5
mild	4	2	4/9	4/5
cool	3	1	3/9	3/5

So here we have 2 input features

• ① outlook

② Temperature

and we have to predict the play $\rightarrow$ yes/no

O/P $\Rightarrow$ play:

play	yes	no
yes	9	
no		5

$$P(\text{yes}) = \frac{9}{14}, \quad P(\text{no}) = \frac{5}{14}$$

Test data $\Rightarrow$ (Sunny, Hot) $\rightarrow$ find O/P

Soln

$$P(\text{yes} / \text{Sunny, Hot}) = \frac{P(\text{yes}) \times P(\text{Sunny} / \text{yes}) \times P(\text{Hot} / \text{yes})}{P(\text{Sunny}) \times P(\text{Hot})}$$

$\downarrow$

I am removing this because it is common factor for all the prediction.

$$\Rightarrow \frac{9}{14} \times \frac{2}{9} \times \frac{2}{9}$$

$$\Rightarrow \frac{1}{7} \times \frac{2}{9} = \frac{2}{63} = \underline{\underline{0.031}}$$

Now :-

$$P(\text{No} / (\text{sunny, Hot})) = P_x(\text{No}) \times P(\text{sunny} / \text{No}) \times P(\text{Hot} / \text{No})$$

$$\Rightarrow \frac{8}{147} \times \frac{3}{8} \times \frac{2}{5}$$

$$= \frac{3}{35} = \underline{\underline{0.085}}$$

$$\Rightarrow P(\text{yes} / (\text{sunny, Hot})) = \underline{\underline{0.031}}$$

$$(0.031 + 0.085)$$

$$\Rightarrow 0.27 \Rightarrow \underline{\underline{27\%}}$$

$$\Rightarrow P(\text{No} / (\text{sunny, Hot})) = \underline{\underline{0.085}}$$

$$(0.031 + 0.085)$$

$$\Rightarrow 0.73 = \underline{\underline{73\%}}$$

Ans { So the output is No because
the chances is greater than yes

Variants of Naive Bayes

① Bernoulli Naive Bayes

Whenever your features are following a Bernoulli distribution, then we need to use Bernoulli Naive Bayes

⇒ Outcomes is ⇒ Pass or fail.

Dataset

f_1	f_2	f_3	O/P
yes	fail	male	yes
no	pass	female	no
yes	pass	female	no
no	fail	male	yes

⇒ Apply the Bernoulli Naive Bayes for these types of data.

② Multinomial Naive Bayes

Whenever you input features are in text format then you have to apply the multinomial Naive Bayes.

dataset : Spam classification.

(Not Spam)
↑
(Spam)

I/P → Email body	O/P (Spam/Ham)
you ^{have} won the million \$ \$ lottery	Spam
You have done good job	Ham (not Spam)

③ Gaussian Naive Bayes.

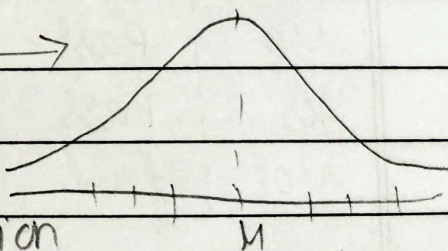
Whenever the features are following the Gaussian distribution then we use the Gaussian Naive Bayes.

dataset



Continuous.

Bell →
Structure
Normal
distribution



↓
{ IRI S }
{ Dataset }